Vidya Sagar Reddy Venna

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ML / Software Engineer — end-to-end ML systems: LLM/RAG pipelines, model serving & MLOps, signal & image processing

EDUCATION

University of Southern California Los Angeles, CA
M.S., Electrical & Computer Engineering (Machine Learning & Data Science) *Aug 2024 – May 2026*
Coursework: Machine Learning, Deep Learning, Image Processing, Neurotechnology Design

IIIT Sri City Sri City, India
B.Tech (Honours), Electronics & Communication Engineering *Aug 2020 – May 2024*
Coursework: DSP, Data Structures & Algorithms, Pattern Recognition, Statistical Signal Processing, Embedded Systems

SKILLS

Languages: Python, C++, SQL, MATLAB, R
ML / AI: PyTorch, TensorFlow, JAX, scikit-learn, CNNs, XGBoost, time-series, OpenCV
LLM / GenAI: RAG, LangChain, Hugging Face, PEFT, evaluation harnesses
MLOps & Serving: FastAPI, Flask, Docker, Kubernetes, MLflow, ONNX, CI/CD, Git, Linux
Data & Cloud: AWS, GCP, Azure, Spark, Kafka
Software: Data Structures, Algorithms, OOP, REST APIs

EXPERIENCE

COOR Lab, USC — Clinical Biomechanics & Sports Research Los Angeles, CA
Graduate Research Assistant *Jul 2025 – Present*
- Built a Python signal-processing pipeline for multi-session patient surface EMG (2 kHz, 7 muscles, 54 of 78 patients) — filtering, TKEO onset detection, and MVIC normalization — producing a quality-controlled dataset used to track post-injury rehabilitation.
- Quantified neuromuscular recovery with linear mixed-effects models and leave-one-out cross-validation, showing force and timing recovery are statistically independent despite strong symptom improvement — informing how clinicians track rehab progress; submitted to APTA CSM 2027.

BrainSightAI Bangalore, India
Machine Learning Intern *Jan 2024 – Jun 2024*
- Ran and validated a Python diffusion-MRI pipeline, computing tract-wise diffusion metrics and benchmarking denoising methods (SNR, RMSE, MSE) to test the reliability of brain-tractography outputs used in clinical imaging.
- Validated dMRI workflows end-to-end (preprocessing, diffusion modeling, tractography) across clinical and public datasets, surfacing sources of variability that affect diagnostic reliability.

Human Movement Analysis Lab, IIIT Sri City Sri City, India
Honours Student Researcher *May 2022 – May 2024*
- Designed and led collection of 4 sEMG datasets (EMAHA-DB4–DB7, 10–25 subjects each) with Noraxon sensors, varying posture, arm position, pace, and load to study ADL and strength-training variability.
- Built end-to-end pipelines (filtering, segmentation, normalization, feature extraction) and trained CNN-BiLSTM and classical ML classifiers; released all 4 datasets publicly, yielding 3 conference publications.

PROJECTS

Scientific Knowledge Extraction — *Information Sciences Institute (ISI)* GitHub
- Built an end-to-end LLM pipeline that extracts structured knowledge (entities, attributes, relationships) from research PDFs, using batched inference and caching to run at scale; added a verification step cross-checking each claim against the source text, reaching 88.3% factual accuracy [95% CI 82.9–93.3].
- Built the evaluation harness — multiple-choice fidelity scoring with bootstrap confidence intervals and an n-gram overlap check to flag copied text.

Predictive Maintenance Platform GitHub
- Built an end-to-end predictive-maintenance platform — remaining-useful-life prediction, bearing-fault detection, model serving (FastAPI), drift monitoring, and automated evaluation — reaching RMSE 16.80 on NASA CMAPSS and PR-AUC 1.00 on CWRU bearing data.

Deep Audio Classification GitHub
- Built a PyTorch pipeline for UrbanSound8K (8.7K clips, 10 classes) comparing MFCC+SVM, VGG-CNN, and ResNet-18 with strict train/test separation (10-fold cross-validation); a Mixup-augmented CNN reached 78.20% ± 6.05% mean accuracy, outperforming all baselines.

State-Space Neural Decoding GitHub
- Implemented and compared Kalman Filter, Point-Process Filter, and PSID models to decode movement from primate neural recordings, evaluating which state-space method best supports brain–computer interface (BCI) applications.

PUBLICATIONS

V. S. R. Venna et al., “Impact of Activity Pace and Arm Position on ADL Classification,” EMBC 2024. [Link]
V. S. R. Venna et al., “Fine-ADL Classification Using sEMG under Varying Conditions,” BIOSIGNALS 2024. [Link]
V. S. R. Venna et al., “Impact of Measurement Conditions on ADL Classification Using Surface EMG,” ISPA 2023. [Link]